



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 04

Objective & Lecture Mapping: In previous labs, you explored Uninformed Search strategies like BFS and UCS inside `agent.py` [cite: 1]. Today, we transition to **Informed Search**. You will enhance your agent by implementing the **A* Search** algorithm, using **Heuristics** to drastically reduce the number of explored nodes in the `visual_grid_game.py` environment [cite: 4]. You will start with basic heuristic functions and connect them to your search logic.

Part 1: Practical Implementation — Building the A* Agent

Step 1.1: Implementing the Heuristic Functions

Informed search algorithms require a heuristic function, $h(n)$, which estimates the cheapest path from the current node to the goal. You will calculate distance metrics on a 2D grid.

1. Create a new Branch called `Week_04` in your Forked Github Repo.
2. Open `agent.py` and locate your `SearchAgent` class [cite: 1].
3. Add a new method named `def manhattan_distance(self, pos, goal):` that calculates the distance using the formula $h(n) = |x_1 - x_2| + |y_1 - y_2|$ and returns the integer value.
4. Add a second method named `def euclidean_distance(self, pos, goal):` that calculates the straight-line distance using the formula $h(n) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$. You will need to `import math` for this.
5. *Testing Checkpoint:* Print the outputs of both functions for a mock start position (0, 0) and goal (3, 4) to verify that Manhattan returns 7 and Euclidean returns 5.0.

Step 1.2: Implementing A* Search

A* evaluates nodes by combining the path cost so far, $g(n)$, and the estimated cost to the goal, $h(n)$. The evaluation function is $f(n) = g(n) + h(n)$.

1. Inside `SearchAgent`, create a new method: `def astar_search(self, start_pos, goal_pos, walls, grid_size, heuristic_type='manhattan')`.
2. Initialize an empty priority queue using `heapq` and an empty set for `reached_states`.
3. Push the initial state into the priority queue. Unlike UCS which only uses $g(n)$, your A* tuple must be formatted as: `(f_cost, g_cost, current_pos, path_taken)`. For the starting node, $g(n) = 0$. Calculate $h(n)$ using your chosen heuristic method, and set $f(n) = g(n) + h(n)$.
4. Create your standard `while` loop to process the queue. Pop the node with the lowest `f_cost`. If `current_pos == goal_pos`, return the `path_taken`. Otherwise, add `current_pos` to `reached_states`.
5. For node expansion, check the four adjacent cells (Up, Down, Left, Right). For each valid neighbor (not a wall, within bounds, and not reached): Calculate $g_{\{new\}} = g_{\{current\}} + 1$, $h_{\{new\}}$ using your heuristic, and $f_{\{new\}} = g_{\{new\}} + h_{\{new\}}$. Push the new tuple to the priority queue.

Step 1.3: Integrating A* into the Agent's Decision Loop

Finally, you need to connect your new search algorithm to the agent's perception and action loop so it can run in the visual environment.

1. Navigate to the `sense_and_act(self, percept)` method in your `SearchAgent`.
2. Add an `elif self.active_algo == 'AStar':` block to handle the new algorithm alongside your existing BFS/DFS/UCS.
3. Extract the necessary global state from the percept dictionary (e.g., `percept['remaining_food']`).
4. Find the closest food item to act as the `goal_pos`. Call your `astar_search` method and store the returned list of actions in `self.plan`.
5. Open `visual_grid_game.py` and modify the `GridGameGUI` initialization to inject your `SearchAgent()` instead of the `ModelBasedAgent()`. Run the simulation and observe the optimized pathfinding!

Part 2: Theoretical Evaluation

Based on your implementation and Lecture 05, complete the following analytical questions.

1. (Understand) What is the key difference between how Uniform-Cost Search (UCS) and A* Search prioritize which node to explore next?

UCS picks the next node with the lowest $g(n)$, the cost so far, so it spreads out evenly in all directions. A* picks the lowest $f(n) = g(n) + h(n)$, cost so far plus estimated cost left. That steers it toward the goal, so it explores far fewer nodes.

2. (Analyze) In Step 1.1, you used Manhattan Distance. Why is Manhattan Distance considered an "admissible" heuristic for this specific 4-way movement grid, and what would happen to your A* algorithm if the heuristic was NOT admissible?

An admissible heuristic never overestimates the true cost ($h(n) \leq h^*(n)$). With only Up/Down/Left/Right moves, each move changes x or y by just 1. So you need at least $|x_1 - x_2| + |y_1 - y_2|$ moves, and walls can only make the real path longer, never shorter. That means Manhattan distance is always equal to or less than the real cost. If the heuristic overestimated, A* could skip the true shortest path because it looked too expensive. It would still find a path, but not necessarily the shortest one, so it would lose its optimality guarantee.

3. (Evaluate) If we modified `visual_grid_game.py` to allow the agent to move diagonally (8-way movement), would Manhattan distance still be an admissible heuristic? Why or why not? Which metric should you switch to?

No. With diagonal moves, one step changes both x and y . Going from $(0,0)$ to $(1,1)$ takes 1 move, but Manhattan says 2, so it overestimates and is no longer admissible. Switch to Chebyshev distance, $\max(|x_1 - x_2|, |y_1 - y_2|)$, which is exact when a diagonal step costs 1. Euclidean distance also never overestimates, so it's admissible too, but it's a weaker estimate, so A* explores more nodes.

4. (Create) When targeting multiple food items simultaneously, calculating the distance to just the single closest food item is a weak heuristic.

Propose (in text) a stronger heuristic for navigating the grid to eat ALL remaining food efficiently.

- $h(n) = (\text{distance to the nearest food}) + (\text{total length of a minimum spanning tree connecting all the remaining food})$.
- The tree is the shortest set of links that joins every food item. The agent must at least reach one food item and then travel between all the others, so this never overestimates and stays admissible.
- It's much closer to the real cost of clearing the whole map than "distance to the closest food". This makes A* choose a smart eating order instead of greedily grabbing the nearest pellet.
- A simpler option that's still admissible is the distance to the farthest remaining food.

Once Completed Get a PDF of the completed File and Push into the Week 04 branch in the Github Project

Finalize and Lock Lab 04 Documentation

**Incomplete: {filledCount}/{fields.length} questions answered.
Please provide detailed responses for all questions.**



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